

Exploring Mechanisms of Suicidal Ideation in Adolescents: Structural Equation Modeling Using DirectLiNGAM

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01 Introduction

Challenges in Suicide Prevention -----

The Difficulty of Predicting Suicidal Thoughts:

Various studies have been conducted on the factors associated with suicidal ideation (e.g. Joiner, 2005; O'Connor, 2018). However, the predictive power of these individual risk factors for suicidal ideation is only at about the chance level (Franklin et.al., 2017).

The Difiiculty of Supporting High-Risk Individuals:

Suicide is a highly heterogeneous phenomenon (Tureaki et.al., 2019), and there is no single, universally effective support strategy.

⇒ By conceptualizing the relationships among complexly intertwined factors as a network and elucidating the causal pathways leading to suicidal ideation, which vary across individuals, it becomes possible to develop support strategies tailored to each person.

02 Method

Participants and Procedure -----

Participants were 2,264 university students (1,004 female, 1,221 male, 39 other; mean = 20.78 years). Data were collected via an online quwstionnaire administered by a research agency.

Measurements -----

- (1) **Suicidal Ideation:** It was measured using a single item adapted from the social survey (NHK Broadcasting Culture Research Institute, 2012).
- (2) **Perceived Burdensomeness and Thwarted Belongingness:** They were measured using items adapted from a Japanese translation of the Interpersonal Needs Questionnaire (Aiba et al., 2019).
- (3) **Multidimensional Assessment:** To assess participants' multidimensional characteristics, items from the Nara Prefecture version of the Multidimensional Assessment Scale (Ito et al., 2021) were used without modification.

03 Result & Discussion

DirectLiNGAM Model -----

We constructed a structural equation model using DirectLiNGAM, creating a network diagram linking 12 factors to suicidal ideation (Figure 1).

Hierarchical Clustering (All Participants) -----

Hierarchical clustering based on relative edge contributions identified characteristic factors among individuals with suicidal ideation. High-risk cluster analysis revealed that low self-acceptance, deteriorating friendships, and physical complaints including sleep deprivation significantly contributed to suicidal ideation.

*To extract the characteristic variables for each cluster, we calculated 95% confidence intervals for the mean differences between the target cluster and the other clusters using the bootstrap method. Variables whose confidence intervals did not include zero were identified as the characteristic variables for that cluster.

Discussion -----

Although suicide involves complex interactions among multiple factors, our findings demonstrate five identifiable patterns within this complexity. Rather than a uniform mechanism, suicidal ideation arises through diverse pathways, with different individuals following distinct factor chains. Tailoring educational and psychological interventions to these specific risk factor patterns may significantly enhance prevention effectiveness.

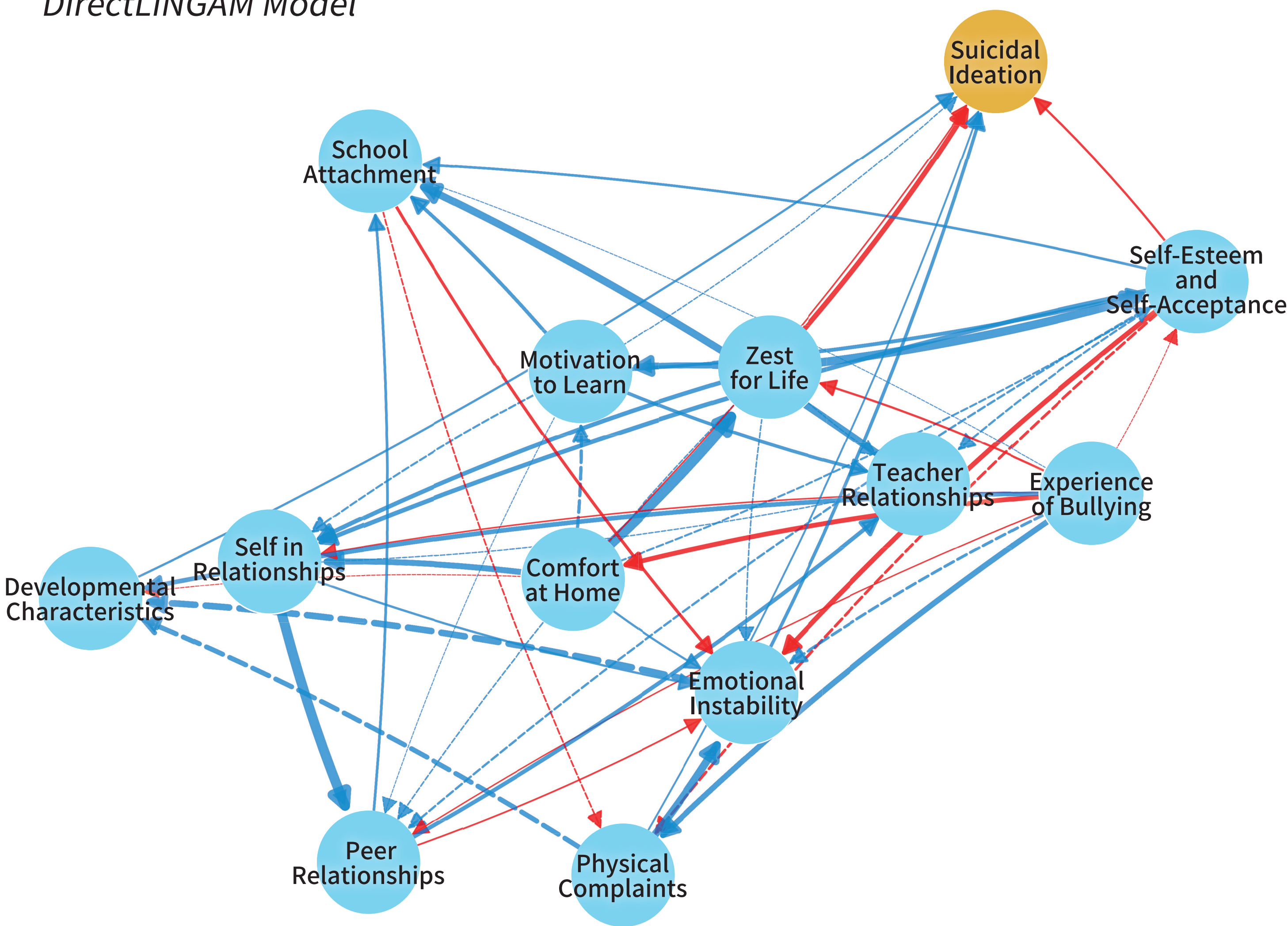
The Purpose of this study -----

- (1) To elucidate the causal pathways leading to suicidal ideation using DirectLiNGAM (Shimizu et.al., 2011)*¹.
- (2) To classify individuals with suicidal ideation into subtypes by focusing on individual differences in causal pathways.

*1: DirectLiNGAM (Shimizu et.al., 2011)

Direct Linear Non-Gaussian Acyclic Model (DirectLiNGAM) is a statistical method that discovers causal relationships from observational data. By assuming the error terms follow a non-Gaussian distribution, it directly estimates a one-way causal network, revealing "what causes what" without requiring experimental manipulation.

Figure 1
DirectLiNGAM Model



Hierarchical Clustering (High Risk Participants) -----

Focusing exclusively on adolescents with frequent suicidal ideation, we performed hierarchical clustering on the network model, identifying six distinct clusters. Five exhibited distinctive factor patterns.

Five Profiles of High Risk Individuals

- (1) Low academic motivation and diminished will to live.
- (2) History of bullying, developmental challenges, emotional instability, and low self-esteem and self-acceptance.
- (3) Somatic symptoms, such as sleep disturbances.
- (4) Family and peer relationship issues, thwarted belongingness, perceived burdensomeness, and diminished will to live.
- (5) Peer relationship difficulties, along with low self-esteem and self-acceptance.